HomeEase: Calculations, VAT and Legal Notes

Study notes from a Claude Code teaching session, 2026-10-17 to 2026-10-18. Source: the repository code and docs/. Pesos are written as PHP.

Not legal or tax advice. These notes describe what the code and docs do. The repo itself says the legal terms await a Philippine lawyer's review.

1. Money basics

- **Rounding:** `roundToCentavo(x) = Math.round(x*100)/100` (backend/src/utils/money.ts). All peso math passes through it.
- **Storage:** bookings and payments store pesos as floats. The ledger uses whole centavos (`toCentavos`) so entries balance exactly.
- **Rates:** commission and withholding rates live in the admin-editable AppSettings row. Environment values are fallbacks only.

2. Job price at booking time

From `computeJobPricing` (utils/pricing.ts) and `taskBasePrice` (services/taskPriceService.ts).

```
basePrice   FIXED: task.basePrice
            PER_UNIT: task.basePrice x quantity (quantity must be >= 1)
            CUSTOM_QUOTE: 0 (worker quotes after inspecting)
distanceFee = max(0, distanceKm - freeDistanceKm) x perKmFee   (defaults 5 km, PHP 10/km)
rushFee     = basePrice x rushFeeRate        (same-day bookings only)
tierFee     = basePrice x (tierMultiplier - 1)
urgencyFee  = 0   (retired, kept for old receipts)
estimatedPrice = basePrice + distanceFee + rushFee + tierFee
```

- Rush and tier fees apply to the base price only, never to distance or add-ons.
- Distance uses the haversine formula (utils/geo.ts), Earth radius 6,371,000 m.
- Example: base 500, 8 km, tier x1.2, no rush. distanceFee = 3 x 10 = 30, tierFee = 100, estimate = 630.

3. Final bill

```
base     = quote exists ? laborCost + materialsCost : estimatedPrice
subtotal = base + approved add-ons
vat      = vatApplicable ? round(subtotal x 0.12) : 0
total    = subtotal + vat + tip
```

4. Commission, withholding and worker payout

```
commission   = round(subtotal x commissionRate)
withholding  = round((subtotal - commission) x withholdingTaxRate)
workerPayout = subtotal - commission - withholding + tip + vat
```

- Rates are snapshotted when the worker accepts the job. A later admin change does not affect it.
- Withholding is taken on the after-commission amount. VAT and tip are never commissioned or taxed.

Example (c=10%, w=5%, subtotal 1,000, VAT on)	Amount (PHP)
Commission	100
Withholding = 5% x 900	45
VAT = 12% x 1,000	120

Example (c=10%, w=5%, subtotal 1,000, VAT on)	Amount (PHP)
Client pays	1,120
Worker receives = 1000 - 100 - 45 + 120	975

5. Settlement and ledger

- **GCash/Maya:** the client pays the full total through a Xendit invoice. The worker is then paid workerPayout.
- **Cash:** the worker keeps the cash. commission + withholding becomes worker debt, covered first by payout credit, then deducted from the next online payout.
- **Ledger:** double-entry, in centavos. Every posting must sum to 0 or it throws. Each posting has a unique idempotency key, so retried webhooks cannot double-post.

```
Online capture:
  XENDIT_CASH               +captured
  COMMISSION_REVENUE        -commission
  WITHHOLDING_TAX_PAYABLE   -withholding
  WORKER_BALANCE            -(invoiced - commission - withholding)
  CLIENT_RECEIVABLE         -overpaid      (only if client overpaid)
```

6. Other rules found

- **Auto-suspend:** totalReviews > 0 and rating <= threshold, or disputes in the last N days >= threshold. Each gate is off when its setting is null.
- **Review strikes:** 3 warnings suspend the account.
- **Not yet studied:** ratings math, tier multipliers, cancellation fees, debt ledger.

7. VAT

Who it belongs to: VAT is the worker's. The client pays it. The platform is neither taxpayer nor remitter.

Party	Role
Client	Pays it. It is added on top of the subtotal.
Worker	The VAT-registered seller. Collects it (included in workerPayout; kept on cash jobs) and files it with the BIR.
App	Computes and records it, shows a summary for the worker's own filing. Never remits it ("The platform never remits this", vatSummaryService.ts).

When it applies

- Only if WorkerProfile.vatRegistered is true. Only an admin can set it, after reviewing the registration.
- Rate is a code constant: VAT_RATE = 0.12. Not admin-editable.
- Frozen on the booking as vatApplicable / vatRate at creation. Settlement never rechecks the worker's live flag.
- Custom-quote tasks freeze it when the worker submits the quote.
- Base is the subtotal only. Tips are excluded (BIR gross-receipts definition). Add-ons, distance fee and mandatory fees are included.

Reporting and threshold

- `generateVatSummaries` sums Payment.vatAmount per worker per period. Informational, for BIR forms 2550Q/2551Q. Platform-funded payments are skipped.
- Hourly sweep sums each unregistered worker's trailing-12-month Payment.subtotal (gross receipts). At PHP 3,000,000 it flags them, notifies worker and admins, writes an audit log. It never sets vatRegistered.

Do not confuse with withholding tax

Withholding tax is deducted from the worker's earnings by the platform, and the platform owes it to the BIR (ledger account WITHHOLDING_TAX_PAYABLE).

Unchecked: whether the platform's own commission is subject to VAT. The code does not compute that.

8. Platform legalities

Worker status

- The worker agreement says HomeEase is an intermediary and does not employ the worker (independent contractor).
- So DOLE minimum wage does not apply by law. constants/doleWageReference.ts is only a pricing sanity check (NCR PHP 755/day, /8 = hourly floor), overridable by an admin with a reason.
- Open question: whether contractor status holds is a labor-law question for a lawyer. The platform sets prices, takes commission and can suspend workers, which a regulator may weigh.

Tax (BIR)

- Withholding is owed to the BIR. Form 2307 PDFs are generated per worker per quarter, with monthly breakdowns.
- Needs the worker's TIN (workers without one are skipped) and AppSettings.atcCode (nothing generates if missing, on purpose).

- The PDF may not match the official BIR template. Have an accountant check it before filing.
- TINs are encrypted in the database and masked in API responses.

Data privacy (RA 10173)

- docs/DATA-MAP.md lists all personal data, where stored, and which outside services receive it (Anthropic, Xendit, Brevo, PhilSMS, Google, Expo, GitHub, Backblaze, Vercel).
- Cross-border transfer is allowed if the platform stays accountable. The Privacy Policy discloses it.
- Sensitive data (IDs, health certificates, clearances, TINs) sits in private buckets with short-lived signed URLs.
- Retention (s.11(e)): expired tokens 1 day, notifications 180 days read / 365 days any, login audit 365 days. Payment, ledger and tax records kept for the legal period.
- User rights: access via a one-time code to the registered email (aim 15 days); deletion is self-service in the app.
- Breach: 72-hour notification plan (docs/INCIDENT-RESPONSE.md) and a yearly NPC incident summary.
- Consent: acceptances stored with version, time, IP and device.

Known open legal gaps (docs/GOVERNANCE.md)

Risk	Gap	Status
R8	Terms not lawyer-reviewed. DTI/SEC business name and address not in the app (Internet Transactions Act).	Open, before public launch
R9	DPO and processing systems not registered with the NPC (due at 1,000+ people with sensitive data).	Open

Controls that support legal claims

- Double-entry, idempotent money records give an audit trail.
- Production DB role cannot edit audit-log or ledger rows.
- Admins require TOTP MFA.

Not found in the code

- BSP / e-money rules (payments go through Xendit).
- Consumer-protection and refund law.
- Background-check law for workers' clearances.
- The platform's own VAT or income tax.

9. Open practice questions

- A PHP 1,000 cash job at 10% commission and 5% withholding, no VAT, no tip: what does the worker owe the platform?
- An unregistered worker completes a PHP 2,000 cash job with a PHP 100 tip (c=10%, w=5%): what do they owe, and what does the client hand over?
- Which of price-setting, commission or suspension power would a labor regulator weigh most on contractor status?

```
Answers to check yourself:
  1 -> commission 100 + withholding 0.05 x 900 = 45  => owes 145
  2 -> commission 200, withholding 0.05 x 1800 = 90   => owes 290; client hands over 2,100
```